

Chapter 6- Virtualization, VMware & Windows Internals

Virtualization (Very Important Concept)

★ What is Virtualization?

👉 Virtualization means creating something **virtual (not physically existing)**

👉 You can **see and use it**, but it doesn't exist as separate hardware

🧠 Core Idea:

👉 Normally:

- 1 Hardware = 1 OS

👉 With Virtualization:

- 1 Hardware = Multiple OS
-

💡 Example:

👉 Like using **multiple mobile apps in one phone** 📱

👉 Or running **Windows inside Windows**

🎯 Why Virtualization?

- Testing OS 🧪
 - Learning IT Support 🖥️
 - Safe environment (no risk to main system) 🔒
-

Check Virtualization Status

Steps:

1. Press **Ctrl + Shift + Esc**
2. Open **Task Manager**
3. Go to **Performance → CPU**

Check:

- Virtualization = **Enabled / Disabled**

If Disabled:

 Go to BIOS → Enable Virtualization

Important:

 Virtualization **cannot be enabled from OS**

 Only from BIOS (Advanced settings)

Virtual Machine Setup using VMware Workstation

What is VMware?

 Software to create virtual machines

Step-by-Step VM Creation

Step 1: Open VMware

 Click **Create New Virtual Machine**

◆ Step 2: Select Installation Method

👉 Choose:

✓ ISO File

👉 Provide path of:

Windows 11

◆ Step 3: Select OS Type

- Microsoft Windows
 - Version: Windows 11 x64
-

◆ Step 4: Name & Location

👉 Give:

- VM Name
 - Storage location
-

◆ Step 5: Set Password

👉 Minimum **8 characters**

👉 Why?

- Keeps VM encrypted 🔒
-

◆ Step 6: Allocate Disk Size

👉 Example:

- 64 GB
-

◆ Step 7: Customize Hardware

👉 Click **Customize Hardware**

Recommended Settings:

- RAM → 4 GB
 - CPU → 4 cores
 - Network Adapter → NAT
-

Why NAT?

 VM and main system are isolated

 If something goes wrong in VM:

✓ Main system is safe

Step 8:

 Click **Finish**

Install Windows inside Virtual Machine

Installation Steps:

Step 1:

 Start VM

 Screen shows:

Press any key to boot

◆ **Step 2:**

👉 Select:

- Language → English
 - Keyboard → Default
-

◆ **Step 3:**

👉 Click:

✓ Install Windows 11

👉 OR:

✓ Repair your PC (for troubleshooting)

◆ **Step 4:**

👉 Accept:

✓ License Agreement

◆ **Step 5:**

👉 Product Key:

✓ Click **I don't have product key**

◆ **Step 6:**

👉 Select Version:

✓ Windows 11 Pro

👉 Why?

- Used in companies
-

◆ **Step 7:**

👉 Select:

✓ Custom Installation

Partitioning Inside Virtual Machine

Scenario:

👉 Total Disk = 64 GB

◆ **Step 1:**

👉 Create partitions:

- 32 GB
 - 32 GB
-

◆ **Step 2:**

👉 System automatically creates:

- 200 MB (System)
 - 16 MB (MSR Reserved)
-

◆ **Step 3:**

👉 Format only **Primary partitions**

✗ Do NOT format:

- System partition
 - MSR partition
-

★ Why Format?

👉 Converts:

- FAT32 → NTFS

👉 NTFS required for Windows

◆ Step 4:

👉 Select partition → Click Next

👉 Installation starts

📁 Windows Folder Structure (VERY IMPORTANT)

📁 Location:

👉 C Drive → Windows

◆ Important Folders:

📁 1. Windows Folder

👉 Contains:

- System files
 - Drivers
 - Fonts
-

📁 2. Program Files

👉 Installed applications stored here

3. ProgramData

 Shared data for all users

Important:

 Many malware hide here

4. Users Folder

 Contains:

- Desktop
 - Downloads
 - Documents
-

5. AppData (Very Important 🔥)

 Location:

C:\Users\Username\AppData

Why Important?

 Malware commonly hides here

Reason:

 Apps run without admin permission

Example:

 You download file → run it

 It may install silently in AppData

Malware Concept (Very Important)

Why AppData is Dangerous?

 Because:

- No admin permission required
 - Auto-execution possible
-

IT Support Tip:

 Always check:

- ✓ AppData folder
 - ✓ Unknown files
-

Important File Locations

1 Notepad Location

 C:\Windows\System32\notepad.exe

2 Fonts Folder

 C:\Windows\Fonts

3 Drivers Location

 C:\Windows\System32\drivers

4 Event Viewer Logs

 C:\Windows\System32\winevt\Logs

Google Chrome Extensions

👉 C:\Users\Username\AppData\Local\Google\Chrome\User Data\Default\Extensions

Troubleshooting Mindset

Scenario-Based Thinking:

VM Not Starting

👉 Check:

- Virtualization enabled
 - BIOS settings
-

Slow VM

👉 Increase:

- RAM
 - CPU cores
-

Malware Issue

👉 Check:

- AppData
 - ProgramData
-



Real IT Support Skills from This Chapter

- ✓ Virtualization setup
 - ✓ OS installation practice
 - ✓ File system understanding
 - ✓ Malware detection basics
 - ✓ Folder-level troubleshooting
-

Final Understanding

- 👉 Virtualization = Multiple OS in one system
 - 👉 VMware = Practice tool
 - 👉 AppData = Malware hotspot
 - 👉 Partitioning = Disk control
-

Chapter 6 (SUMMARY)

Virtualization (Core Concept)

👉 Virtualization = Virtual environment create karna

👉 1 system → multiple OS run

💡 Example:

- Windows inside Windows
- Testing environment

Why Virtualization?

- Testing OS 
- Learning IT Support 
- Safe environment 

Virtualization Enable Check

👉 Task Manager → CPU → Virtualization

⚠️ If Disabled:

👉 BIOS me enable karna padega

VMware (Virtual Machine Tool)

👉 VMware = Software to create VM

VM Setup (Important)

- RAM → 4GB
- CPU → 4 cores

- Disk → 50–64GB
- Network → NAT 🌐

👉 NAT = Isolation (safe environment)

Windows Installation in VM

👉 Same steps as real system:

- Boot from ISO
- Partition
- Format NTFS
- Install OS

Windows Folder Structure (VERY IMPORTANT)

Windows Folder

👉 System files

Program Files

👉 Installed apps

ProgramData

👉 Shared data

⚠️ Malware hide here

Users Folder

👉 User data

AppData

 Most important

 Malware hotspot:

- Hidden folder
 - No admin permission required
-

Malware Concept

 Malware hides in:

- AppData
- ProgramData

 Reason:

- Easy execution
 - Hidden location
-

Important System Locations

- Notepad → System32
 - Drivers → System32\drivers
 - Logs → Event Viewer
-

Troubleshooting Concepts

 VM not starting:

- Check virtualization

 Slow VM:

- Increase RAM/CPU

 Malware:

- Check AppData
-

CONCLUSION

 Core learning:

- Virtualization = Practice safely
- VMware = Real-world tool
- AppData = Security risk
- Folder knowledge = troubleshooting power

 Final line:

“A smart IT Support engineer doesn’t just fix system, he understands what is happening inside” 

Q & A

◆ VIRTUALIZATION QUESTIONS

? Q1: What is virtualization?

👉 Answer:

Virtualization is the process of creating a virtual version of a computer system, allowing multiple operating systems to run on a single physical machine.

? Q2: Why virtualization is used?

👉 Answer:

- Testing OS
 - Learning
 - Safe environment
 - Running multiple OS
-

? Q3: Can virtualization be enabled from OS?

👉 Answer:

✗ No

👉 It can only be enabled from BIOS settings

◆ VMWARE QUESTIONS

? Q4: What is VMware?

👉 Answer:

VMware is a virtualization software used to create and run virtual machines on a physical system.

? Q5: What are basic requirements for VM?

👉 Answer:

- RAM → Minimum 4GB
 - CPU → Multi-core
 - Disk → 50GB+
-

? Q6: What is NAT in VMware?

👉 Answer:

NAT (Network Address Translation) allows VM to access internet while keeping it isolated from host system.

◆ WINDOWS INTERNALS QUESTIONS

? Q7: What is AppData folder?

👉 Answer:

AppData is a hidden folder that stores application data for users.

? Q8: Why AppData is important?

👉 Answer:

Because many applications and malware store data here without requiring admin permission.

? Q9: What is ProgramData?

👉 Answer:

ProgramData stores shared application data accessible by all users.

? Q10: What is System32 folder?

👉 Answer:

System32 contains critical system files, drivers, and core Windows components.

◆ MALWARE QUESTIONS (VERY IMPORTANT)

? Q11: Where does malware usually hide?

👉 Answer:

- AppData
 - ProgramData
 - Temporary folders
-

? Q12: Why malware hides in AppData?

👉 Answer:

Because:

- No admin permission required
 - Hidden from user
 - Easy to execute
-

◆ TROUBLESHOOTING QUESTIONS

VM NOT STARTING

? Q13: VM not starting, what will you do?

👉 Answer:

1. Check virtualization enabled
 2. Check BIOS settings
 3. Restart system
-

SLOW VM

? Q14: VM running slow?

👉 Answer:

- Increase RAM
 - Increase CPU cores
 - Close background apps
-

MALWARE ISSUE

? Q15: System infected with malware?

👉 Answer:

- Check AppData
 - Check unknown processes
 - Run antivirus
 - Remove suspicious files
-

◆ REAL JOB SCENARIOS

? **Q16: User says system behaving strangely?**

👉 **Answer:**

- Check background processes
 - Check AppData
 - Scan system
-

? **Q17: Software not working properly?**

👉 **Answer:**

- Check Program Files
 - Check dependencies
 - Reinstall application
-

? **Q18: How do you practice OS installation safely?**

👉 **Answer:**

- 👉 Use Virtual Machine
 - 👉 Avoid using real system
-

◆ INTERVIEW IMPRESSION QUESTIONS

? **Q19: Why virtualization is important for IT support?**

👉 **Answer:**

It allows safe testing, learning, and troubleshooting without affecting the main system.

? Q20: How do you detect malware manually?

👉 Answer:

- Check AppData
 - Check unknown files
 - Monitor processes
-

IKM